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Reg. No. :

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Question Paper Code : 30028

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2023.

Third Semester

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Artificial Intelligence and Data Science

AL 3391 — ARTIFICIAL INTELLIGENCE

(Regulations 2021)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

- For each of the environments below determine what type of agent architecture is most appropriate (table lookup, simple reflex, goal-based or utility-based).
 - Medical Diagnosis System
 - Satellite Image Analysis System
 - Part Picking Robot
 - Interactive English Tutor
- Formulate PEAS for an automated taxi driver.
- Compare and Contrast Admissible and Consistent Heuristics.
- Comment on the statement "Breadth-first search is a special case of uniform-cost search".
- Class scheduling: There is a fixed number of professors and classrooms, a list of classes to be offered, and a list of possible time slots for classes. Each professor has a set of classes that he or she can teach. Give a precise formulation.
- Justify why we cannot use traditional min max for games with an element of chance, such as backgammon.
- If the unicorn is mythical, then it is immortal, but if it is not mythical, it is a mortal mammal. If the unicorn is either immortal or a mammal, then it is horned. The unicorn is magical if it is horned. Prove: the unicorn is horned by resolution.

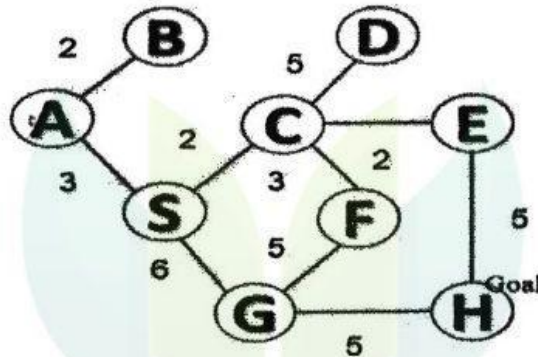
8. Some people like every vegetable. Convert it to First order Logic.
9. Justify the purpose of Bayesian networks.
10. What is Causal networks?

PART B — ($5 \times 13 = 65$ marks)

11. (a) Consider the water jug problem: You are given two jugs, a 4-gallon one and 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallon of water from the 4-gallon jug? Explain the state space representation and apply an optimal sequence of actions to solve it.

Or

(b)



Perform BFS, DFS, Uniform cost search strategies on the following graph and also formulate the algorithm for any 2 strategies.

12. (a) Elaborate on the need for Local search algorithms and discuss any one algorithm in detail.

Or

- (b) Discuss on Online Search Agents that uses depth-first exploration.

13. (a) Brief on MinMax algorithm and also discuss the need for alpha beta pruning.

Or

- (b) Formulate the Map-coloring problem as CSP and explain with an example.

14. (a) (i) Brief on the concept of Resolution and explain the Propositional Resolution Algorithm. (5)
- (ii) Prove the following axioms using the Resolution Algorithm. (8)
- (1) All hounds howl at night.
 - (2) Anyone who has any cats will not have any mice.
 - (3) Light sleepers do not have anything which howls at night.
 - (4) John has either a cat or a hound.
 - (5) (Conclusion) If John is a light sleeper, then John does not have any mice.

Or

- (b) Give the rules of inferences in Propositional Logic. (8)
- Which rule of inference is used in each argument below? (5)
- (i) Alice is a Math major. Therefore, Alice is either a Math major or a CSI major.
 - (ii) Jerry is a Math major and a CSI major. Therefore, Jerry is a Math major.
 - (iii) If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.
 - (iv) If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
 - (v) If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.
15. (a) Design a Bayesian belief network for the diagnosis of car's electrical system.

Or

- (b) Briefly explain exact inference in Bayesian Networks.

PART C — (1 × 15 = 15 marks)

16. (a) The missionaries and cannibals problem is usually stated as follows. Three missionaries and three cannibals are on one side of a river, along with a boat that can hold one or two people. Find a way to get everyone to the other side without ever leaving a group of missionaries in one place outnumbered by the cannibals in that place.
- (i) Formulate the problem precisely, making only those distinctions necessary to ensure a valid solution. Draw a diagram of the complete state space. (5)

Question Paper Code : 50069

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2024.

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Third/Fourth Semester

Artificial Intelligence and Data Science

AL 3391 – ARTIFICIAL INTELLIGENCE

(Common to: Computer Science and Engineering (Artificial Intelligence and Machine Learning))

(Regulations 2021)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. There are well-known classes of problems that are intractably difficult for computers and other classes that are provably undecidable. Does this mean that AI is impossible?
2. Formulate PEAS for an automated Bill paying system.
3. What is admissible heuristics?
4. Give two examples of partially observable environments.
5. Given two jugs of capacities 5 litres and 3 litres with no measuring markers on them. Assume that there is endless supply of water. What is the minimum number of states to measure 4 litres of water?
6. What for Monte-Carlo tree search is used?
7. How do you represent "All dogs have tails" in First order logic?
8. What are knowledge –based agents? Name its parts.
9. Two students A and B have registered for a certain course. Student A attends the class 80% of the time. Student B attends the class 60% of the time. Assuming their absences are independent, what is the probability that both neither show up to class on any given day?
10. What are causal networks?

PART B — (5 × 13 = 65 marks)

11. (a) (i) List some environmental properties that are important for intelligent agents. (7)
 (ii) Describe the properties of the environment for the Mars Rover. (6)

Or

- (b) Consider the search graph shown below in Fig. 11(b). S is the start state and G is the goal state. All edges are bidirectional. For each of the following search strategies, give the path that would be returned and the pseudocode for the same.
 (i) Depth-first graph search (4)
 (ii) Breadth-first graph search (4)
 (iii) Uniform cost graph search (5)

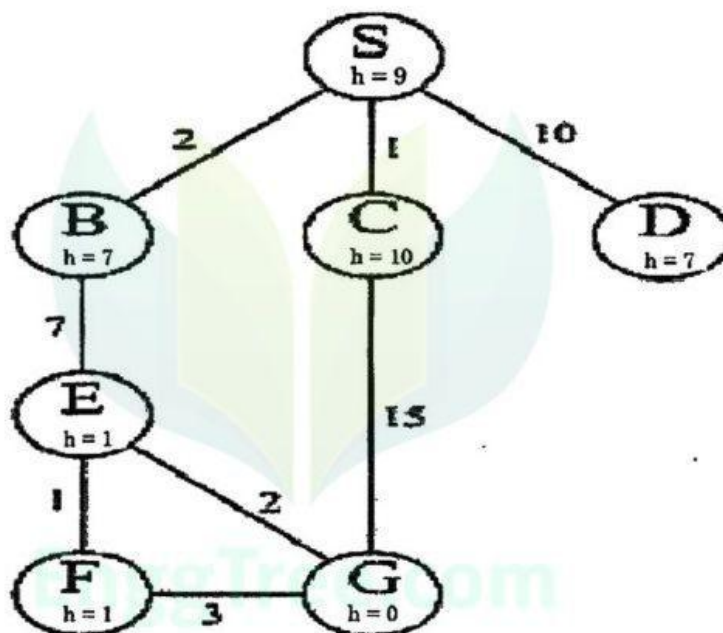


Fig. 11(b).

12. (a) Discuss on any two Local Search algorithms with examples.

Or

- (b) Discuss on online search agents with example.

13. (a) Describe how the minimax and alpha-beta algorithms change for two-player, non-zero-sum games in which each player has a distinct utility function and both utility functions are known to both players. If there are no constraints on the two terminal utilities, is it possible for any node to be pruned by alpha-beta? What if the player's utility functions on any state differ by at most a constant k , making the game almost cooperative?

Or

- (b) Consider the so-called Crypt-arithmetic problem shown below.

SAVE+
 MORE
 ———
 MONEY

The problem is to figure out the digits (0 to 9) that should be assigned to these letters so that the indicated arithmetic operation is valid. Your job is to formulate this as a search problem. Clearly indicate a definition of state, initial state, goal state, operators and define a suitable path cost. Estimate the size of the search tree and draw a small segment of the search tree.

14. (a) Discuss on Knowledge representation and engineering.

Or

- (b) Differentiate backward and forward chaining with example.
15. (a) We have a bag of three biased coins a, b, and c with probabilities of coming up heads of 30%, 60%, and 75%, respectively. One coin is drawn randomly from the bag (with equal likelihood of drawing each of the three coins), and then the coin is flipped three times to generate the outcomes X1, X2, and X3.
- (i) Draw the Bayesian network corresponding to this setup and define the necessary Conditional Probability Tables. (7)
- (ii) Calculate which coin was most likely to have been drawn from the bag if the observed flips come out heads twice and tails once. (6)

Or

- (b) How does Exact inference differ from approximate Inference? Give example.

PART C — (1 × 15 = 15 marks)

16. (a) The traveling salesperson problem (TSP) can be solved with the minimum-spanning-tree (MST) heuristic, which estimates the cost of completing a tour, given that a partial tour has already been constructed. The MST cost of a set of cities is the smallest sum of the link costs of any tree that connects all the cities.
- (i) Show how this heuristic can be derived from a relaxed version of the TSP. (3)
- (ii) Show that the MST heuristic dominates straight-line distance. (3)
- (iii) Write a problem generator for instances of the TSP where cities are represented by random points in the unit square. (4)
- (iv) Find an efficient algorithm for constructing the MST and use it with a graph search to solve instances of the TSP. (5)

Or

(b) CSP: Air Traffic Control

We have five planes: A, B, C, D, and E and two runways: international and domestic. We would like to schedule a time slot and runway for each aircraft to either land or take off. We have four time slots: {1, 2, 3, 4} for each runway, during which we can schedule a landing or take off of a plane. We must find an assignment that meets the following constraints:

- Plane B has lost an engine and must land in time slot 1.
- Plane D can only arrive at the airport to land during or after time slot 3.
- Plane A is running low on fuel but can last until at most time slot 2.
- Plane D must land before plane C takes off, because some passengers must transfer from D to C.
- No two aircrafts can reserve the same time slot for the same runway.

Brief on CSP and complete the formulation of this problem as a CSP in terms of variables, domains and constraints (both unary and binary). Constraints should be expressed implicitly using mathematical or logical notation rather than with words.

Question Paper Code : 70016

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2022.

Third Semester

Artificial Intelligence and Data Science

AL3391 – ARTIFICIAL INTELLIGENCE

(Regulations 2021)

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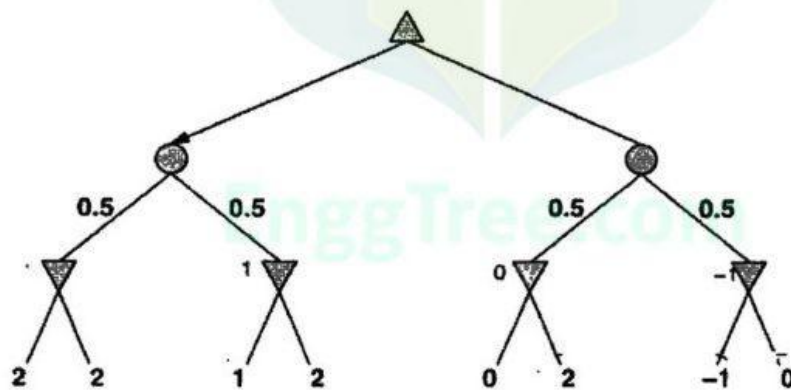
Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Characterize the environment of an agent playing soccer.
2. Can an agent that senses only partial information about the state always be perfectly rational?
3. Define admissible and dominant heuristics.
4. What is the purpose of a contingency plan?
5. In the game tree given below, Δ represents Max node, ∇ represents min node and $\circ$ represents chance nodes. Find the utility value of the root node.



6. Give a precise formulation of the following constraint satisfaction problem in terms of variables, domain and constraints: There are a five professors and 10 classrooms, a list of classes to be offered, and a list of possible time slots for classes Each professor has a set of classes that he or she can teach.

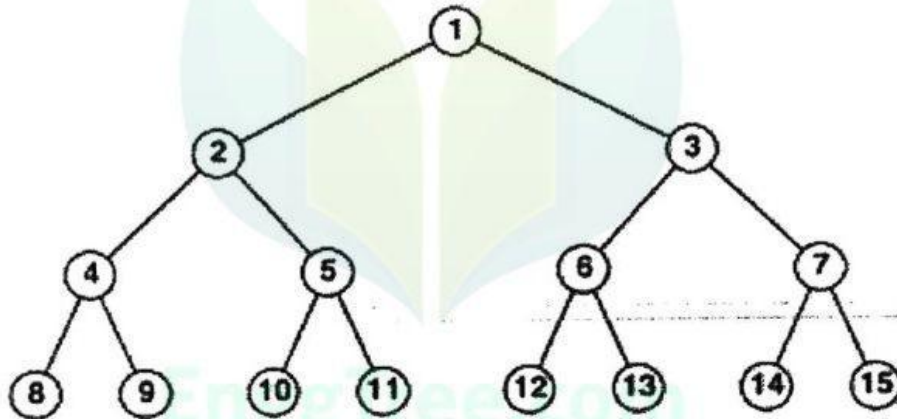
7. State the converse and contrapositive of the statement "When I stay up late, it is necessary that I sleep until noon".
8. Define the terms belief state and state estimation.
9. How are Bayesian networks represented?
10. What is the purpose of relational probability models?

PART B — ($5 \times 13 = 65$ marks)

11. (a) Define an agent. Explain the four basic agents that embody the principles underlying intelligent systems with examples.

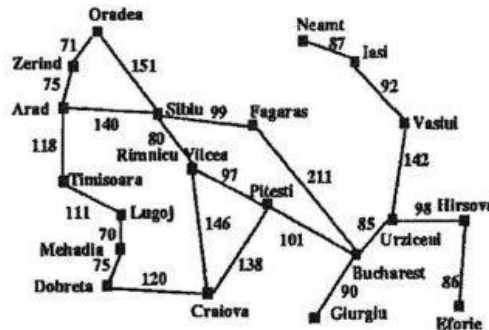
Or

- (b) Consider a state space given below.



- (i) If the goal state is numbered '11', list the order in which the states will be visited using (1) breadth first search (2) depth first search (3) depth limited search with a limit 2 (depth of the root is 0) (4) iterative deepening search.
- (ii) If iterative deepening search is used to traverse the state space, how many times will the state space tree be constructed to reach the goal?
- (iii) If bidirectional search is used to reach the goal state, what would be the branching factor in the forward and backward direction?

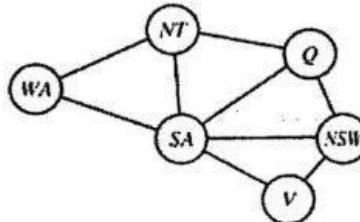
12. (a) Outline A* Algorithm. Trace the algorithm to find the shortest route from Lugoj to Bucharest using the straight line distance heuristic. Show the sequence of nodes traversed by the algorithm. The straight line distances are given below.



Town	Air Dist.	Town	Air Dist.
Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
Dobreta	242	Pitesti	100
Eforie	161	Rimnicu Vilcea	193
Fagaras	176	Sibiu	253
Giurgiu	77	Timisoara	329
Hirsova	151	Urziceni	80
Iasi	226	Vaslui	199
Lugoj	244	Zerind	374

Or

- (b) How does online agents search in unknown environments?
13. (a) Consider the map coloring problem with 6 variables and three colors (red, green and blue). The constraint graph for the problem is given below. How does backtracking search solve the given problem? What are the heuristics used to improve the efficiency of the search? How are failures detected early in backtracking? Can Breadth first search be applied to the above problem? State reasons for your answer.



Or

- (b) What are partially observable games? How are they solved in a deterministic environment?

14. (a) Express the following statements as predicates:

- All people who are not poor and are smart are happy
- Those people who read are not stupid.
- John can read and is wealthy.
- Happy people have exciting lives.

Skolemize the above statements if required and prove the following by resolution: "Can anyone be found with an exciting life?" (Assume $\sim$ stupid $\equiv$ smart, wealthy $\equiv \sim$ poor)

Or

- (b) During a murder investigation, you have gathered some clues. Express them as propositions and solve the following scenario using laws of inference. The clues gathered are given below:

- If the knife is in the store room, then we saw it when we cleared the store room.
- The murder was committed at the basement or inside the apartment.
- If the murder was committed at the basement, then the knife is in the yellow dust bin.
- We did not see a knife when we cleared the store room.
- If the murder was committed outside the building, then we are unable to find the knife;
- If the murder was committed inside the apartment, then the knife is in the store room.

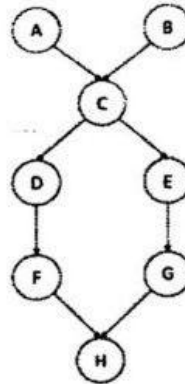
Find: "where is the knife?"

15. (a) (i) Consider there are 3 boolean variables toothache, catch and cavity. From the full joint distribution given below, calculate the following: (8)

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

- (1) $P(\text{toothache})$
- (2) $P(\text{Cavity})$
- (3) $P(\text{Toothache} \mid \text{cavity})$
- (4) $P(\text{Cavity} \mid \text{toothache} \vee \text{catch})$

- (ii) What is d-separation? When are two nodes d-separated? From the Bayesian network given below, find whether D and E d-separated given evidence about both A and B? State reasons for your answer. (5)

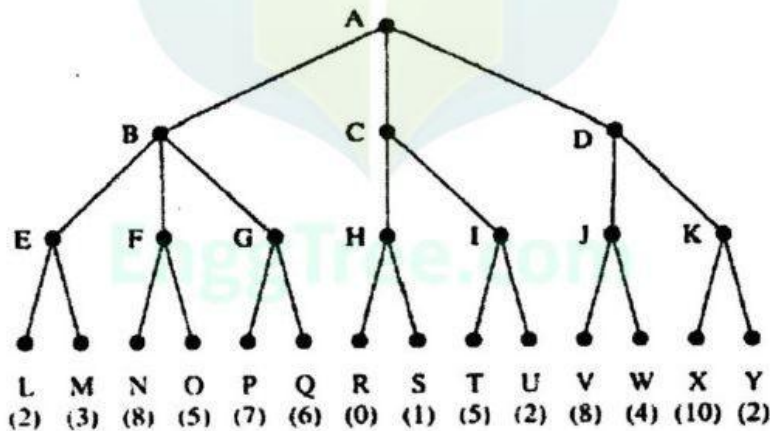


Or

- (b) How does direct sampling methods help in approximate inference?

PART C — (1 × 15 = 15 marks)

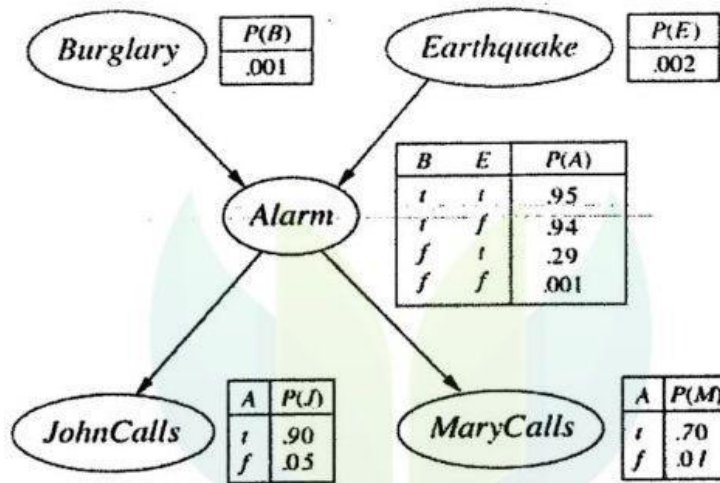
16. (a) Consider the following game tree in which the static scores (in parentheses at the tip nodes) are all from the first players point of view. Assume that the first player is the maximizing player.



Which move should the first payer choose? Use minimax algorithm on the game tree and list the nodes which would not be examined using the alpha-beta algorithm assuming that nodes are examined in left-to-right order? Will the same branches be pruned if the nodes are examined In right-to-left order? Is alpha-beta algorithm guaranteed to force a win whenever possible? State reasons.

Or

- (b) You have a new burglar alarm installed at home. It is fairly reliable at detecting a burglary, but also responds on occasion to minor earthquakes. You also have two neighbors, John and Mary, who have promised to call you at work when they hear the alarm. John nearly always calls when he hears the alarm, but sometimes confuses the telephone ringing with the alarm and calls then, too. Mary, on the other hand, likes rather loud music and often misses the alarm altogether. The Bayesian network and the conditional probability table (CPT) for the scenario is given below. In the CPTs, the letters B, E, A, J, and M stand for Burglary, Earthquake, Alarm, JohnCalls, and Marycalls, respectively. From the Bayesian network, find $P(B/J,M)$ using variable elimination.



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